Grokking Dynamic Programming Patterns for Coding Interviews

Master Dynamic Programming by learning coding patterns. Act smartly, and follow the Dynamic Programming patterns.

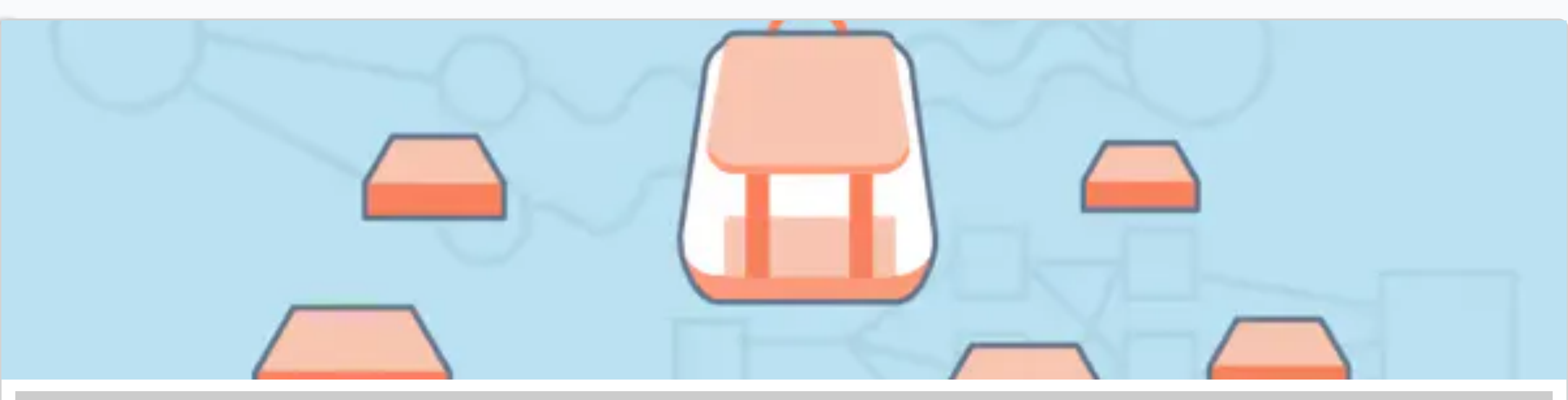
Level: Advanced | Study Time: 71h | Lessons: 73 | Playgrounds : 142

39,106 (Students)

Course Overview

The fact is, Dynamic Programming (DP) problems can be some of the most intimidating on a coding interview. Even when it's actually clear if a problem can be solved using DP (which it rarely is), it can be pretty challenging to even know where to start on the solution. Unless, that is, you're trained on the approach to solving DP problems.

This course was made from scratch with just that goal in mind. It'll equip you with a set of easy-to-understand techniques to handle any DP problem. Rather than just having you try to memorize solutions, you'll be walked through five underlying DP patterns that can then be applied to solve 35+ DP problems. In each pattern, we'll start with a recursive brute-force solution -- this is actually the best way to start solving a DP problem! Once we have a recursive solution, we'll then apply the advanced DP methods of Memoization and Tabulation. The practice problems in this course were carefully chosen, covering the most frequently asked DP questions in coding interviews.



Start Reading

Certification on completing the course.

Download Certificate

Course Content

1. Introduction
1 chapter

- What is Dynamic Programming?

2. Pattern 1: 0/1 Knapsack
13 chapters

- Introduction
- 0/1 Knapsack
- Solution: 0/1 Knapsack
- Equal Subset Sum Partition
- Solution: Equal Subset Sum Partition
- Subset Sum
- Solution: Subset Sum
- Minimum Subset Sum Difference
- Solution: Minimum Subset Sum Difference
- Count of Subset Sum
- Solution: Count of Subset Sum
- Target Sum
- Solution: Target Sum

3. Pattern 2: Unbounded Knapsack
10 chapters

- Unbounded Knapsack
- Solution: Unbounded Knapsack
- Rod Cutting
- Solution: Rod Cutting
- Coin Change
- Solution: Coin Change
- Minimum Coin Change
- Solution: Minimum Coin Change
- Maximum Ribbon Cut
- Solution: Maximum Ribbon Cut

4. Pattern 3: Fibonacci Numbers
12 chapters

- Fibonacci numbers
- Solution: Fibonacci numbers
- Staircase
- Solution: Staircase
- Number factors
- Solution: Number factors
- Minimum jumps to reach the end
- Solution: Minimum jumps to reach the end
- Minimum jumps with fee
- Solution: Minimum jumps with fee
- House thief
- Solution: House thief

5. Pattern 4: Palindromic Subsequence
10 chapters

- Longest Palindromic Subsequence
- Solution: Longest Palindromic Subsequence
- Longest Palindromic Substring
- Solution: Longest Palindromic Substring
- Count of Palindromic Substrings
- Solution: Count of Palindromic Substrings
- Minimum Deletions in a String to make it a Palindrome
- Solution: Minimum Deletions in a String to make it a Palindrome
- Palindromic Partitioning
- Solution: Palindromic Partitioning

6. Pattern 5: Longest Common Substring
26 chapters

- Longest Common Substring
- Solution: Longest Common Substring
- Longest Common Subsequence
- Solution: Longest Common Subsequence
- Minimum Deletions & Insertions to Transform a String into another
- Solution: Minimum Deletions & Insertions to Transform a String into another
- Longest Increasing Subsequence
- Solution: Longest Increasing Subsequence
- Maximum Sum Increasing Subsequence
- Solution: Maximum Sum Increasing Subsequence
- Shortest Common Super-sequence
- Solution: Shortest Common Super-sequence
- Minimum Deletions to Make a Sequence Sorted
- Solution: Minimum Deletions to Make a Sequence Sorted
- Longest Repeating Subsequence
- Solution: Longest Repeating Subsequence
- Subsequence Pattern Matching
- Solution: Subsequence Pattern Matching
- Longest Bitonic Subsequence
- Solution: Longest Bitonic Subsequence
- Longest Alternating Subsequence
- Solution: Longest Alternating Subsequence
- Edit Distance
- Solution: Edit Distance
- Strings Interleaving
- Solution: Strings Interleaving

7. Appendix
1 chapter

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